# Coordination-Dependent Hydrogen Adsorption on Co-, Fe-, and Ni-Functionalized Monolayer $CrCl_3$

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# Supporting Information Available

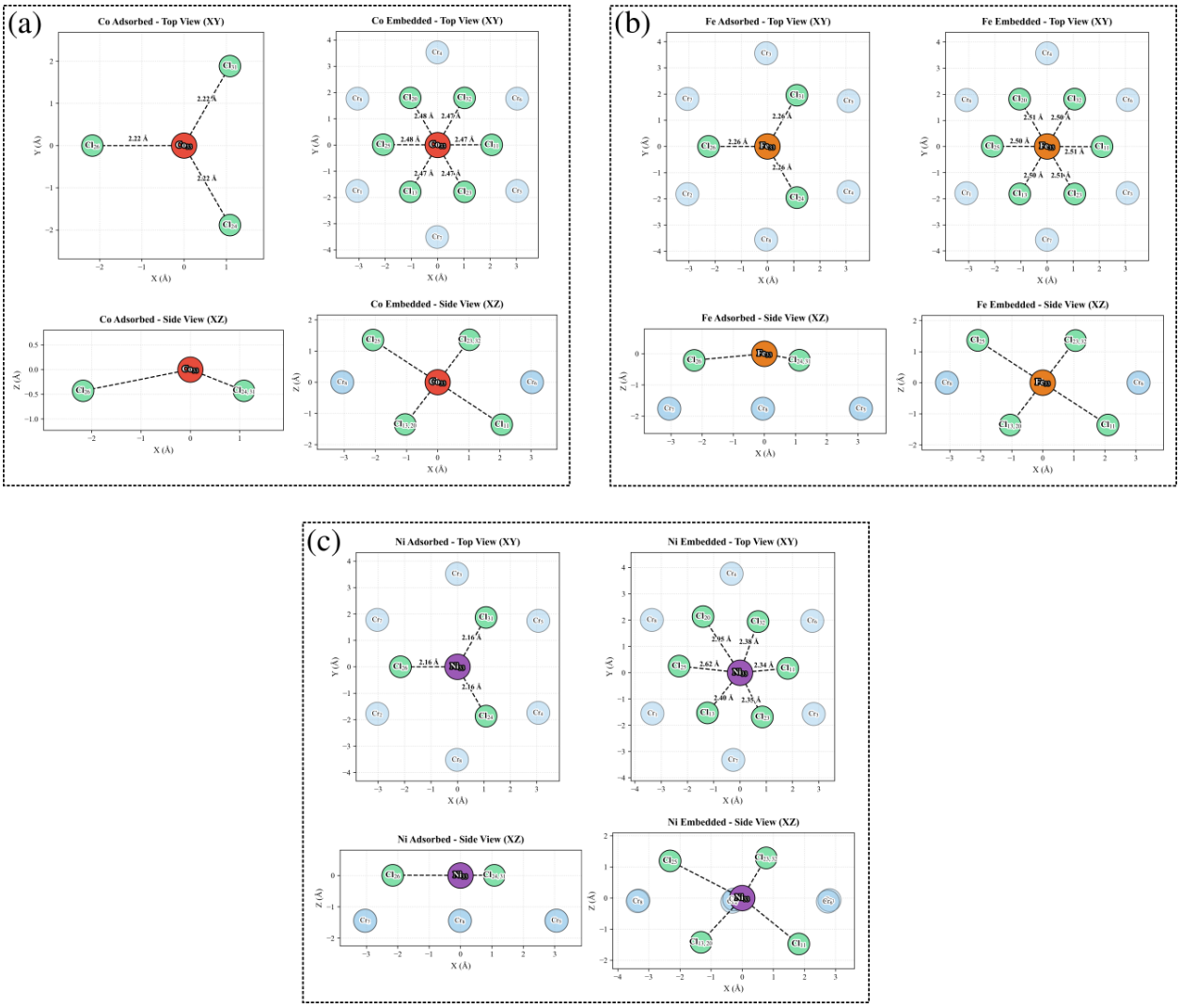



Fig. S1. Schematic top ($XY$) and side ($XZ$) views of the optimized local coordination geometries and TM–Cl bond lengths (Å) for (a) Co-, (b) Fe-, and (c) Ni-functionalized monolayer $CrCl_3$ in both adsorbed (hollow site, 3-fold coordinated) and embedded (6-fold octahedrally coordinated) configurations. Light blue, light green, and red/orange/purple spheres represent Cr, Cl, and transition-metal (Co, Fe, Ni) atoms, respectively.

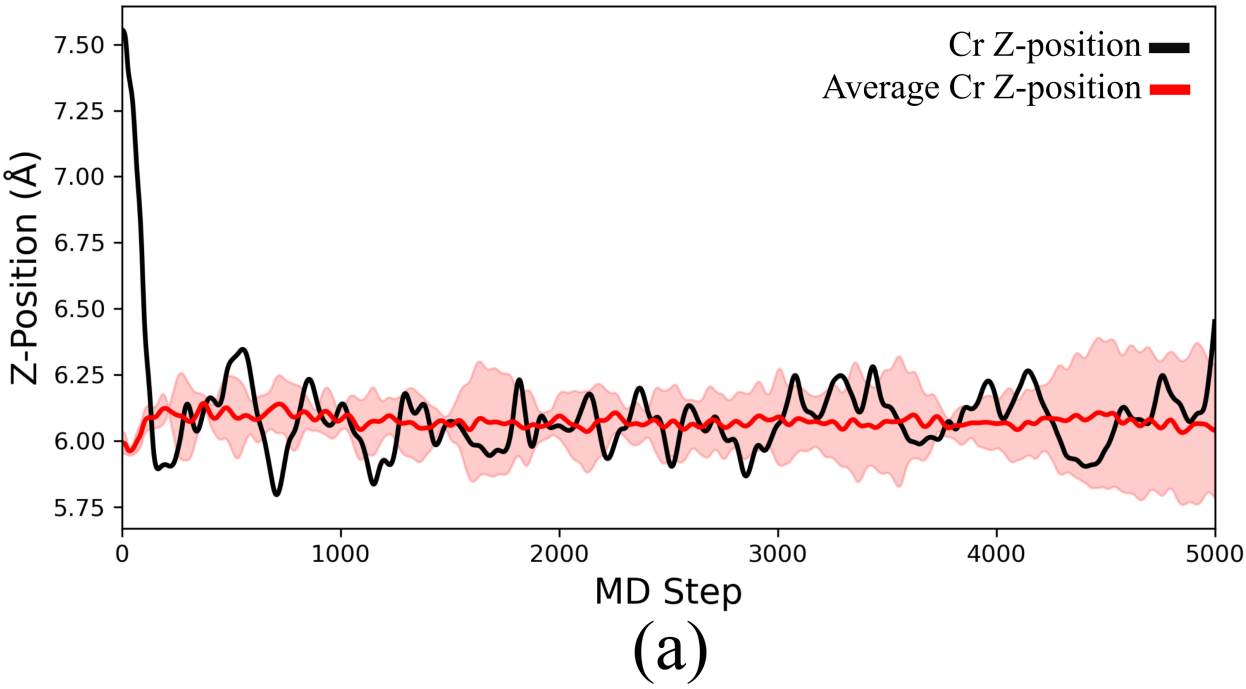



Fig. S2. Time evolution of the absolute $z$ coordinate of the Cr atom initially adsorbed at the hollow site of monolayer $CrCl_3$ during a 5 ps *ab initio* molecular dynamics simulation at 300 K. The black curve represents the instantaneous $z$ coordinate of the adsorbed Cr atom, whereas the red curve denotes the average $z$ coordinate of the Cr atoms belonging to the host monolayer.

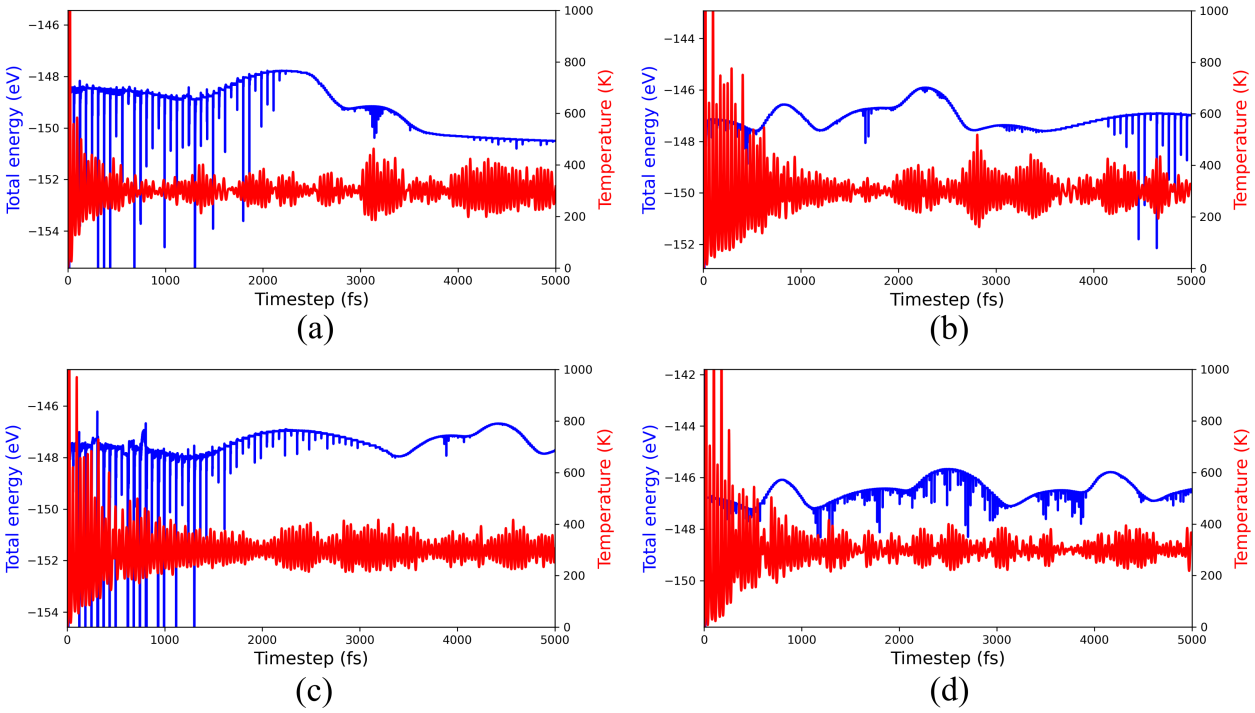



Fig. S3. Time evolution of the total energy (blue curves, left axes) and instantaneous temperature (red curves, right axes) during the 5 ps AIMD simulations at 300 K for monolayer $CrCl_3$ containing an initially surface-adsorbed (a) Cr, (b) Co, (c) Fe, or (d) Ni atom at the hollow site. Following the initial thermostatting transients, the temperatures fluctuate around the target value. The changes in the total-energy profiles accompany the structural rearrangements sampled during the trajectories.

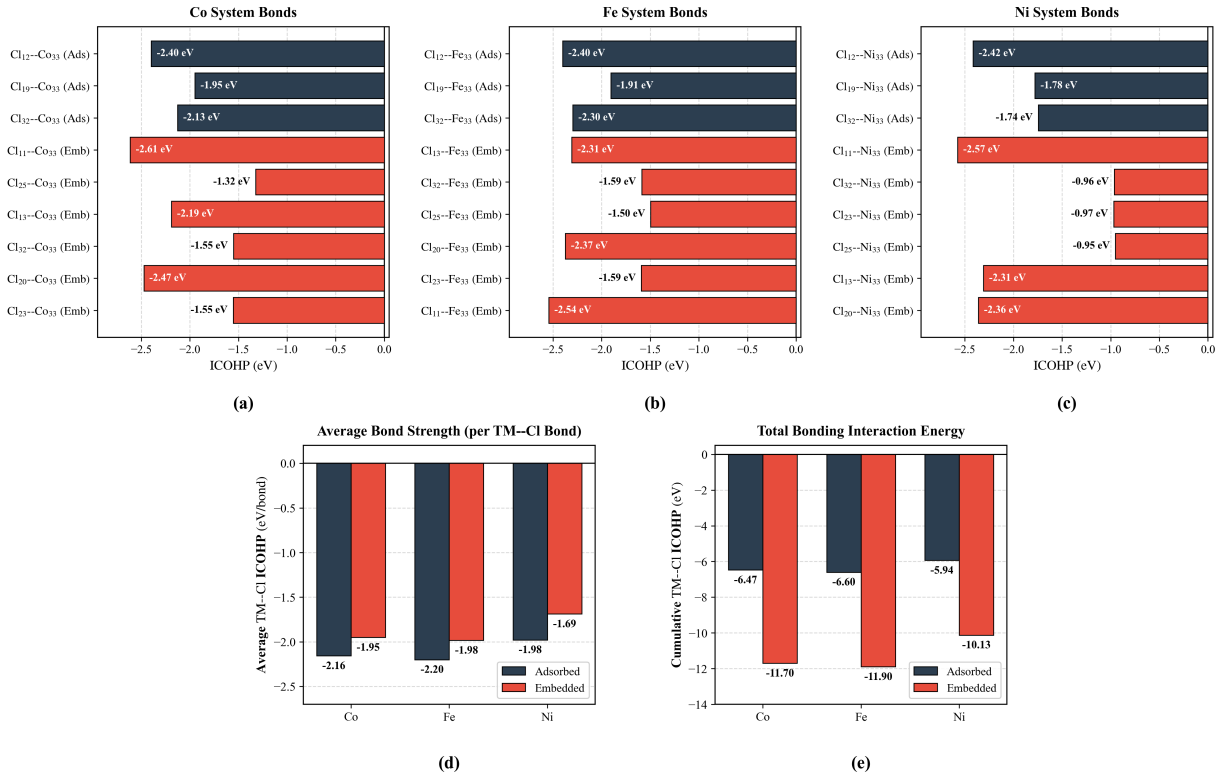



Fig. S4. TM–Cl chemical-bonding analysis and relative stability of surface-adsorbed and embedded transition-metal configurations in monolayer $CrCl_3$ under the multi-site $+U_{\mathrm{all}}$ tier ($U_{\mathrm{Cr}} = U_{\mathrm{TM}} = 3.29$ eV; consistent with Figure 6, with full Pure PBE and $+U_{\mathrm{all}}$ comparisons consolidated in Table S4). (a–c) Individual integrated crystal orbital Hamilton population (ICOHP) values for the Co–Cl, Fe–Cl, and Ni–Cl interactions, respectively. Dark navy bars correspond to the threefold coordinated surface-adsorbed configurations at the $S_2$ hollow site, whereas red bars correspond to the sixfold coordinated embedded configurations. More negative ICOHP values indicate stronger net bonding interactions. (d) Average ICOHP per TM–Cl bond for the adsorbed and embedded configurations. (e) Cumulative TM–Cl ICOHP obtained by summing the individual ICOHP contributions over the corresponding coordination shell.

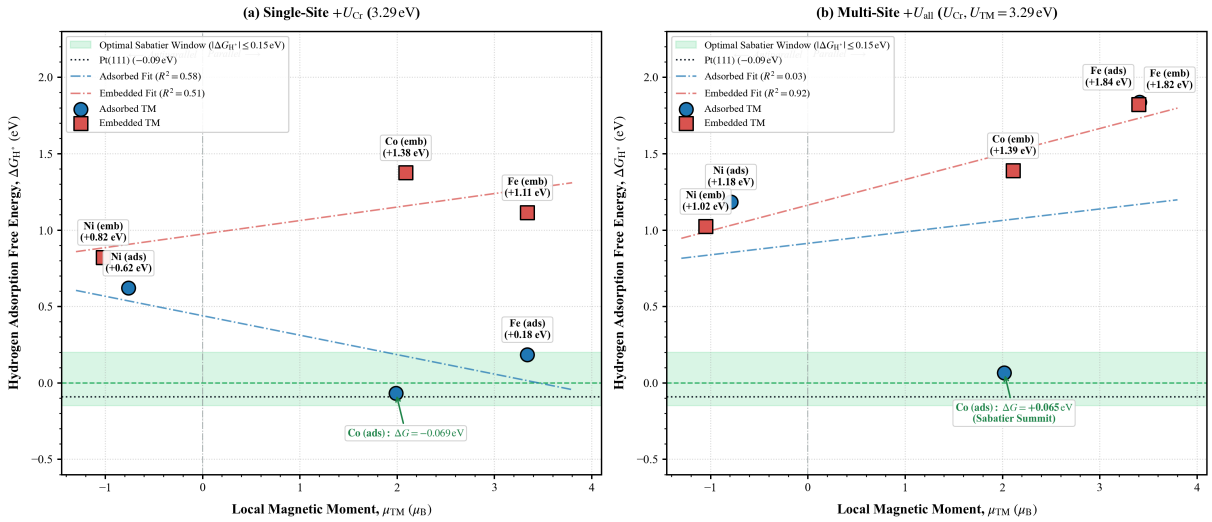



Fig. S5. Relationship between the local magnetic moment of the functionalizing transition-metal atom, $\mu_{\mathrm{TM}}$, in the relaxed H-free configurations and the hydrogen adsorption free energy, $\Delta G_{\mathrm{H}^*}$, across two systematic Hubbard $U$ tiers: (a) single-site $+U_{\mathrm{Cr}}$ ($U_{\mathrm{Cr}} = 3.29$ eV) and (b) multi-site $+U_{\mathrm{all}}$ ($U_{\mathrm{Cr}} = U_{\mathrm{TM}} = 3.29$ eV). Blue circles and red squares represent the surface-adsorbed ($S_2$) and embedded transition-metal configurations in monolayer $CrCl_3$, respectively. The vertical dash-dotted line at $\mu_{\mathrm{TM}} = 0$ separates antiparallel (Ni) from parallel (Co, Fe) magnetic coupling relative to the ferromagnetic Cr host. Shaded green bands denote the optimal Sabatier catalytic window ($|\Delta G_{\mathrm{H}^*}| \leq 0.15$ eV) with the Pt(111) benchmark ($-0.09$ eV) indicated by dotted horizontal lines. Magnetic moments are expressed in units of the Bohr magneton, $\mu_{\mathrm{B}}$.

# Supplementary Tables: Comprehensive Structural, Energetic, and Thermodynamic Benchmarks

## Table S1: Multi-Scale Hydrogen Adsorption on Pristine Monolayer $CrCl_3$

Table S1. Comprehensive multi-scale evaluation of hydrogen adsorption on pristine monolayer $CrCl_3$ across three supercell dimensions ($1\times1$, $2\times2$, and $3\times3$), three high-symmetry adsorption sites (S1: Top-Cl, S2: Hollow, S3: Top-Cr), and three levels of exchange-correlation dispersion treatment (pure PBE, PBE+D3 zero-damping, and PBE+D3 Becke-Johnson damping). All calculations were performed spin-polarized with $E_{\text{cut}} = 400$ eV and commensurate Monkhorst-Pack $k$-point grids ($10\times10\times1$, $5\times5\times1$, and $3\times3\times1$, respectively). Adsorption energies are referenced to isolated $H_2$ gas: $\Delta E_{\text{ads}} = E_{\text{host+H}} - E_{\text{host}} - \frac{1}{2}E(H_2)$. Free energies are evaluated as $\Delta G_{\text{ads}} = \Delta E_{\text{ads}} + \Delta\text{ZPE} - T\Delta S$, where explicit phononic columns ($\Delta$ZPE and $-T\Delta S$) are provided for complete vibrational frequency assignment.

| Supercell | Site | Coord. | Method / vdW | $E_{\text{host}}$ (eV) | $E_{\text{host+H}}$ (eV) | $\Delta E_{\text{ads}}$ (eV) | $\Delta\text{ZPE}^*$ (eV) | $-T\Delta S^*$ (eV) | $\Delta G_{\text{ads}}$ (eV) | $\mu_{\text{tot}}$ ($\mu_B$) |
|---|---|---|---|---|---|---|---|---|---|---|
| $1\times1$ ($\theta = 1.00$) | S1 | Top-Cl | PBE (no vdW) | −39.0717 | −40.7734 | +1.678 | +0.040 | +0.200 | +1.918 | 7.00 |
| | S2 | Hollow | PBE (no vdW) | −39.0717 | −39.9647 | +2.487 | +0.040 | +0.200 | +2.727 | 5.04 |
| | S3 | Top-Cr | PBE (no vdW) | −39.0717 | −40.8155 | +1.636 | +0.040 | +0.200 | +1.876 | 5.00 |
| | S1 | Top-Cl | PBE+D3 (zero) | −39.9841 | −41.8232 | +1.542 | +0.040 | +0.200 | +1.782 | 7.00 |
| | S2 | Hollow | PBE+D3 (zero) | −39.9841 | −40.9868 | +2.378 | +0.040 | +0.200 | +2.618 | 5.51 |
| | S3 | Top-Cr | PBE+D3 (zero) | −39.9841 | −41.7093 | +1.656 | +0.040 | +0.200 | +1.896 | 5.00 |
| | S1 | Top-Cl | PBE+D3 (BJ) | −40.5268 | −42.5826 | +1.325 | +0.040 | +0.200 | +1.565 | 7.00 |
| | S2 | Hollow | PBE+D3 (BJ) | −40.5268 | −41.8481 | +2.060 | +0.040 | +0.200 | +2.300 | 7.00 |
| | S3 | Top-Cr | PBE+D3 (BJ) | −40.5268 | −42.2405 | +1.667 | +0.040 | +0.200 | +1.907 | 5.00 |
| $2\times2$ ($\theta = 0.25$) | S1 | Top-Cl | PBE (no vdW) | −156.2868 | −158.0877 | +1.579 | +0.040 | +0.200 | +1.819 | 25.00 |
| | S2 | Hollow | PBE (no vdW) | −156.2868 | −157.1957 | +2.471 | +0.040 | +0.200 | +2.711 | 23.00 |
| | S3 | Top-Cr | PBE (no vdW) | −156.2868 | −158.0422 | +1.624 | +0.040 | +0.200 | +1.864 | 23.00 |
| | S1 | Top-Cl | PBE+D3 (zero) | −159.9366 | −161.8036 | +1.514 | +0.040 | +0.200 | +1.754 | 25.00 |
| | S2 | Hollow | PBE+D3 (zero) | −159.9366 | −160.9571 | +2.361 | +0.040 | +0.200 | +2.601 | 23.00 |
| | S3 | Top-Cr | PBE+D3 (zero) | −159.9366 | −161.6926 | +1.625 | +0.040 | +0.200 | +1.865 | 23.00 |
| | S1 | Top-Cl | PBE+D3 (BJ) | −162.1074 | −164.2008 | +1.288 | +0.040 | +0.200 | +1.528 | 25.00 |
| | S2 | Hollow | PBE+D3 (BJ) | −162.1074 | −163.1641 | +2.324 | +0.040 | +0.200 | +2.564 | 23.01 |
| | S3 | Top-Cr | PBE+D3 (BJ) | −162.1074 | −163.8525 | +1.636 | +0.040 | +0.200 | +1.876 | 23.00 |
| $3\times3$ ($\theta = 0.11$) | S1 | Top-Cl | PBE (no vdW) | −351.6468 | −353.7608 | +1.266 | +0.040 | +0.200 | +1.506 | 55.00 |
| | S2 | Hollow | PBE (no vdW) | −351.6468 | −352.5562 | +2.470 | +0.040 | +0.200 | +2.710 | 53.00 |
| | S3 | Top-Cr | PBE (no vdW) | −351.6468 | −353.4262 | +1.600 | +0.040 | +0.200 | +1.840 | 53.00 |
| | S1 | Top-Cl | PBE+D3 (zero) | −364.7418 | −366.8243 | +1.298 | +0.040 | +0.200 | +1.538 | 55.00 |
| | S2 | Hollow | PBE+D3 (zero) | −364.7418 | −365.7984 | +2.324 | +0.040 | +0.200 | +2.564 | 53.00 |
| | S3 | Top-Cr | PBE+D3 (zero) | −364.7418 | −366.4930 | +1.630 | +0.040 | +0.200 | +1.870 | 53.00 |
| | S1 | Top-Cl | PBE+D3 (BJ) | −364.7418 | −366.7699 | +1.353 | +0.040 | +0.200 | +1.593 | 55.00 |
| | S2 | Hollow | PBE+D3 (BJ) | −364.7418 | −365.7988 | +2.324 | +0.040 | +0.200 | +2.564 | 53.00 |
| | S3 | Top-Cr | PBE+D3 (BJ) | −364.7418 | −366.4846 | +1.638 | +0.040 | +0.200 | +1.878 | 53.00 |

*Placeholder zero-point energy ($\Delta\text{ZPE} = +0.040$ eV) and entropic ($-T\Delta S = +0.200$ eV at 298.15 K) values represent the standard computational hydrogen electrode (CHE) reference terms ($\Delta G = \Delta E_{\text{ads}} + 0.24$ eV). These explicit columns are formatted to receive the full explicit phonon/vibrational frequency mode calculations upon completion by the co-authors.

## Table S2: Transition-Metal Functionalization on Monolayer $CrCl_3$ ($2 \times 2$ Supercell)

Table S2. Structural energetics, magnetic states, and relative site stabilities of transition-metal adatoms (Co, Fe, Ni) functionalized on monolayer $CrCl_3$ ($2 \times 2$ supercell). Evaluated across Pure PBE ($E_{\text{host}} = -156.287$ eV), PBE+D3(BJ) ($E_{\text{host}} = -162.107$ eV), and DFT+U ($U_{\text{eff}} = 3.29$ eV applied exclusively to Cr $3d$, $E_{\text{host}} = -147.304$ eV). Interaction energy is $E_{\text{int}} = E_{\text{tot}} - E_{\text{host}}$; relative site penalty is $\Delta E_{\text{rel}} = E_{\text{tot}} - E_{\text{tot,GS}}$. Bond distance $d_{\text{TM-X}}$ reflects coordination to nearest ligand Cl (S1, S2, Emb) or underlying Cr (S3). Ground state configurations within each functional tier are highlighted in bold.

| Theory | Site (Coordination) | $E_{\text{tot}}$ (eV) | $E_{\text{int}}$ (eV) | $\Delta E_{\text{rel}}$ (eV) | $\mu_{\text{tot}}$ ($\mu_{\text{B}}$) | $d_{\text{TM-X}}$ (Å) | Thermodynamic State |
|---|---|---|---|---|---|---|---|
| ***A. Cobalt Functionalization ($CrCl_3$–Co)*** | | | | | | | |
| Pure PBE | S1 (Top-Cl, 1-fold) | −161.015 | −4.728 | **0.000** | 23.00 | 2.14 (Cl) | **Ground State (surface)** |
| | S2 (Hollow, 3-fold) | −161.013 | −4.726 | +0.002 | 23.00 | 2.24 (Cl) | Metastable |
| | S3 (Top-Cr, 3-fold) | −160.366 | −4.079 | +0.649 | 23.00 | 2.59 (Cr) | Metastable |
| | Embedded (6-fold) | −160.804 | −4.517 | +0.211 | 27.12 | 2.31 (Cl) | Embedded pore |
| PBE+D3 (BJ) | S1 (Top-Cl, 1-fold) | −166.942 | −4.835 | +0.222 | 27.00 | 2.13 (Cl) | Metastable |
| | S2 (Hollow, 3-fold) | −167.164 | −5.057 | **0.000** | 23.00 | 2.23 (Cl) | **Ground State (surface)** |
| | S3 (Top-Cr, 3-fold) | −166.454 | −4.347 | +0.710 | 23.00 | 2.58 (Cr) | Metastable |
| PBE+D3+$U$ (3.29) | S1 (Top-Cl, 1-fold) | −152.444 | −5.141 | **0.000** | 27.03 | 2.20 (Cl) | **Ground State (surface)** |
| | S2 (Hollow, 3-fold) | −152.368 | −5.064 | +0.076 | 27.00 | 2.21 (Cl) | Metastable |
| | S3 (Top-Cr, 3-fold) | −151.773 | −4.469 | +0.671 | 27.00 | 2.58 (Cr) | Metastable |
| ***B. Iron Functionalization ($CrCl_3$–Fe)*** | | | | | | | |
| Pure PBE | S1 (Top-Cl, 1-fold) | −161.905 | −5.618 | +0.715 | 28.41 | 2.16 (Cl) | Metastable |
| | S2 (Hollow, 3-fold) | −162.620 | −6.333 | **0.000** | 28.44 | 2.28 (Cl) | **Ground State (surface)** |
| | S3 (Top-Cr, 3-fold) | −161.314 | −5.027 | +1.306 | 21.65 | 2.62 (Cr) | Metastable |
| | Embedded (6-fold) | −162.108 | −5.821 | +0.512 | 28.96 | 2.36 (Cl) | Embedded pore |
| PBE+D3 (BJ) | S1 (Top-Cl, 1-fold) | −168.914 | −6.807 | **0.000** | 22.00 | 2.14 (Cl) | **Ground State (surface)** |
| | S2 (Hollow, 3-fold) | −168.116 | −6.009 | +0.797 | 28.39 | 2.26 (Cl) | Metastable |
| | S3 (Top-Cr, 3-fold) | −165.913 | −3.806 | +3.001 | 24.18 | 2.61 (Cr) | Metastable |
| PBE+D3+$U$ (3.29) | S1 (Top-Cl, 1-fold) | −154.834 | −7.531 | **0.000** | 29.97 | 2.40 (Cl) | **Ground State (surface)** |
| | S2 (Hollow, 3-fold) | −153.430 | −6.126 | +1.405 | 29.29 | 2.25 (Cl) | Metastable |
| | S3 (Top-Cr, 3-fold) | −152.932 | −5.628 | +1.902 | 29.44 | 2.65 (Cr) | Metastable |
| ***C. Nickel Functionalization ($CrCl_3$–Ni)*** | | | | | | | |
| Pure PBE | S1 (Top-Cl, 1-fold) | −159.789 | −3.502 | +0.290 | 24.00 | 2.12 (Cl) | Metastable |
| | S2 (Hollow, 3-fold) | −160.078 | −3.792 | **0.000** | 24.00 | 2.21 (Cl) | **Ground State (surface)** |
| | S3 (Top-Cr, 3-fold) | −159.285 | −2.998 | +0.794 | 24.00 | 2.55 (Cr) | Metastable |
| | Embedded (6-fold) | −159.319 | −3.032 | +0.760 | 24.00 | 2.28 (Cl) | Embedded pore |
| PBE+D3 (BJ) | S1 (Top-Cl, 1-fold) | −165.914 | −3.807 | +0.392 | 24.00 | 2.11 (Cl) | Metastable |
| | S2 (Hollow, 3-fold) | −166.306 | −4.199 | **0.000** | 24.00 | 2.20 (Cl) | **Ground State (surface)** |
| | S3 (Top-Cr, 3-fold) | −165.284 | −3.178 | +1.022 | 24.00 | 2.54 (Cr) | Metastable |
| PBE+D3+$U$ (3.29) | S1 (Top-Cl, 1-fold) | −151.461 | −4.157 | +0.215 | 24.00 | 2.18 (Cl) | Metastable |
| | S2 (Hollow, 3-fold) | −151.676 | −4.372 | **0.000** | 24.00 | 2.16 (Cl) | **Ground State (surface)** |
| | S3 (Top-Cr, 3-fold) | −150.752 | −3.448 | +0.924 | 24.00 | 2.64 (Cr) | Metastable |

## Table S3: Hydrogen Evolution Reaction (HER) Adsorption Thermodynamics on TM-Functionalized $CrCl_3$

Table S3. Comprehensive hydrogen adsorption energetics, Caique system-specific vibrational/entropic corrections, and theoretical HER overpotentials ($\eta_{\mathrm{HER}}$) across pristine, surface-adsorbed, and embedded single-atom catalytic centers on monolayer $CrCl_3$. Energetics are benchmarked across pure PBE, PBE+D3(BJ), and PBE+D3+$U$ ($U_{\mathrm{eff}} = 3.29$ eV on Cr-3$d$). Adsorption free energies are calculated via $\Delta G_{\mathrm{H}^*} = \Delta E_{\mathrm{ads}} + \Delta E_{\mathrm{ZPE}} - T\Delta S$. Theoretical overpotentials are defined by the Volmer-step thermodynamic descriptor: $\eta_{\mathrm{HER}} = |\Delta G_{\mathrm{H}^*}|/e$.

| **Catalyst System** | **Site / Geometry** | $\boldsymbol{\Delta E_{\mathrm{ZPE}}}$ **(eV)** | $\boldsymbol{-T\Delta S}$ **(eV)** | $\boldsymbol{\Delta E_{\mathrm{ZPE}} - T\Delta S}$ **(eV)** | $\boldsymbol{\Delta G_{\mathrm{H}^*}}$ **(eV)** | | | $\boldsymbol{\eta_{\mathrm{HER}}}$ **(V)** |
|---|---|---|---|---|---|---|---|---|
| | | | | | **Pure PBE** | **PBE+D3** | **PBE+D3+U** | **(PBE+D3+U)** |
| ***A. Pristine Monolayer Reference*** | | | | | | | | |
| $CrCl_3$ | Top-Cl ($S_1$) | +0.03 | +0.18 | +0.21 | +1.789 | +1.498 | +1.125 | 1.125 |
| $CrCl_3$ | Hollow ($S_2$) | +0.03 | +0.18 | +0.21 | +2.681 | +2.534 | +2.040 | 2.040 |
| $CrCl_3$ | Top-Cr ($S_3$) | +0.06 | +0.18 | +0.26 | +1.884 | +1.896 | +2.436 | 2.436 |
| ***B. Surface-Adsorbed Transition Metals (3-fold Coordinated Hollow Site)*** | | | | | | | | |
| $CrCl_3$–Co (ads) | Atop Co | +0.05 | +0.19 | +0.24 | +0.178 | +0.178 | **−0.069** | **0.069** |
| $CrCl_3$–Fe (ads) | Atop Fe | +0.01 | +0.17 | +0.18 | +0.315 | +0.375 | **+0.185** | **0.185** |
| $CrCl_3$–Ni (ads) | Atop Ni | +0.02 | +0.17 | +0.19 | +0.812 | +0.862 | +0.620 | 0.620 |
| ***C. Embedded Transition Metals (6-fold Octahedral Pore Site)*** | | | | | | | | |
| $CrCl_3$–Co (emb) | Atop Co | +0.05 | +0.19 | +0.26 | +1.756 | +1.743 | +1.375 | 1.375 |
| $CrCl_3$–Fe (emb) | Atop Fe | +0.07 | +0.19 | +0.26 | +1.348 | +1.367 | +1.114 | 1.114 |
| $CrCl_3$–Ni (emb) | Atop Ni | +0.01 | +0.19 | +0.20 | +1.848 | +0.927 | **+0.821** | **0.821** |

*Vibrational frequencies and thermodynamic entropy were computed via density functional perturbation theory / finite differences (`IBRION = 5`) by co-author Caique Campos de Oliveira. $U_{\mathrm{eff}} = 3.29$ eV was applied to Cr-3$d$ via the Dudarev formulation with Grimme D3(BJ) dispersion.

## Table S4: Quantitative Crystal Orbital Hamilton Population (ICOHP) Bonding Analysis Across Functional Tiers

Table S4. Quantitative Crystal Orbital Hamilton Population (COHP) and integrated ICOHP bonding parameters for surface-adsorbed ($S_2$ hollow) and embedded transition-metal centers (Co, Fe, Ni) in monolayer $CrCl_3$, comparing the Pure PBE baseline against the multi-site $+U_{\mathrm{all}}$ tier ($U_{\mathrm{Cr}} = U_{\mathrm{TM}} = 3.29$ eV; corresponding to Figure **??**). Values are evaluated at $E_{\mathrm{F}}$ using LOBSTER 4.1.0 with Bunge localized atomic basis projections. Individual bond pairs, equilibrium bond lengths ($d_{\mathrm{TM-Cl}}$ in Å), average ICOHP per bond ($\langle$ICOHP$\rangle$ in eV/bond), and cumulative ICOHP ($\sum$ICOHP in eV) are listed.

| System / Site | Coord. | Bond Pairs | $d_{\mathrm{TM-Cl}}$ (Å) | | ⟨ICOHP⟩ (eV/bond) | | $\sum$ICOHP (eV) | |
|---|---|---|---|---|---|---|---|---|
| | | | Pure PBE | $+U_{\mathrm{all}}$ | Pure PBE | $+U_{\mathrm{all}}$ | Pure PBE | $+U_{\mathrm{all}}$ |
| $CrCl_3$–Co (ads) | 3-fold Cl ($S_2$) | $Cl_{12}, Cl_{19}, Cl_{32}$ | 2.22 | 2.21 | −2.23 | **−2.16** | −6.70 | **−6.47** |
| $CrCl_3$–Fe (ads) | 3-fold Cl ($S_2$) | $Cl_{12}, Cl_{19}, Cl_{32}$ | 2.26 | 2.25 | −2.28 | **−2.20** | −6.84 | **−6.60** |
| $CrCl_3$–Ni (ads) | 3-fold Cl ($S_2$) | $Cl_{12}, Cl_{19}, Cl_{32}$ | 2.16 | 2.16 | −2.06 | **−1.98** | −6.18 | **−5.94** |
| $CrCl_3$–Co (emb) | 6-fold Cl (Pore) | $Cl_{11}, Cl_{13}, Cl_{20}, Cl_{23}, Cl_{25}, Cl_{32}$ | 2.47–2.48 | 2.40–2.49 | −1.87 | **−1.95** | −11.22 | **−11.70** |
| $CrCl_3$–Fe (emb) | 6-fold Cl (Pore) | $Cl_{11}, Cl_{13}, Cl_{20}, Cl_{23}, Cl_{25}, Cl_{32}$ | 2.50–2.51 | 2.44–2.50 | −1.90 | **−1.98** | −11.40 | **−11.90** |
| $CrCl_3$–Ni (emb) | 6-fold Cl (Pore) | $Cl_{11}, Cl_{13}, Cl_{20}, Cl_{23}, Cl_{25}, Cl_{32}$ | 2.34–2.95 | 2.36–2.48 | −1.67 | **−1.69** | −10.01 | **−10.13** |

# Figure S6: Multi-Level Functional Benchmark and Caique Vibrational Corrections for HER Free Energies

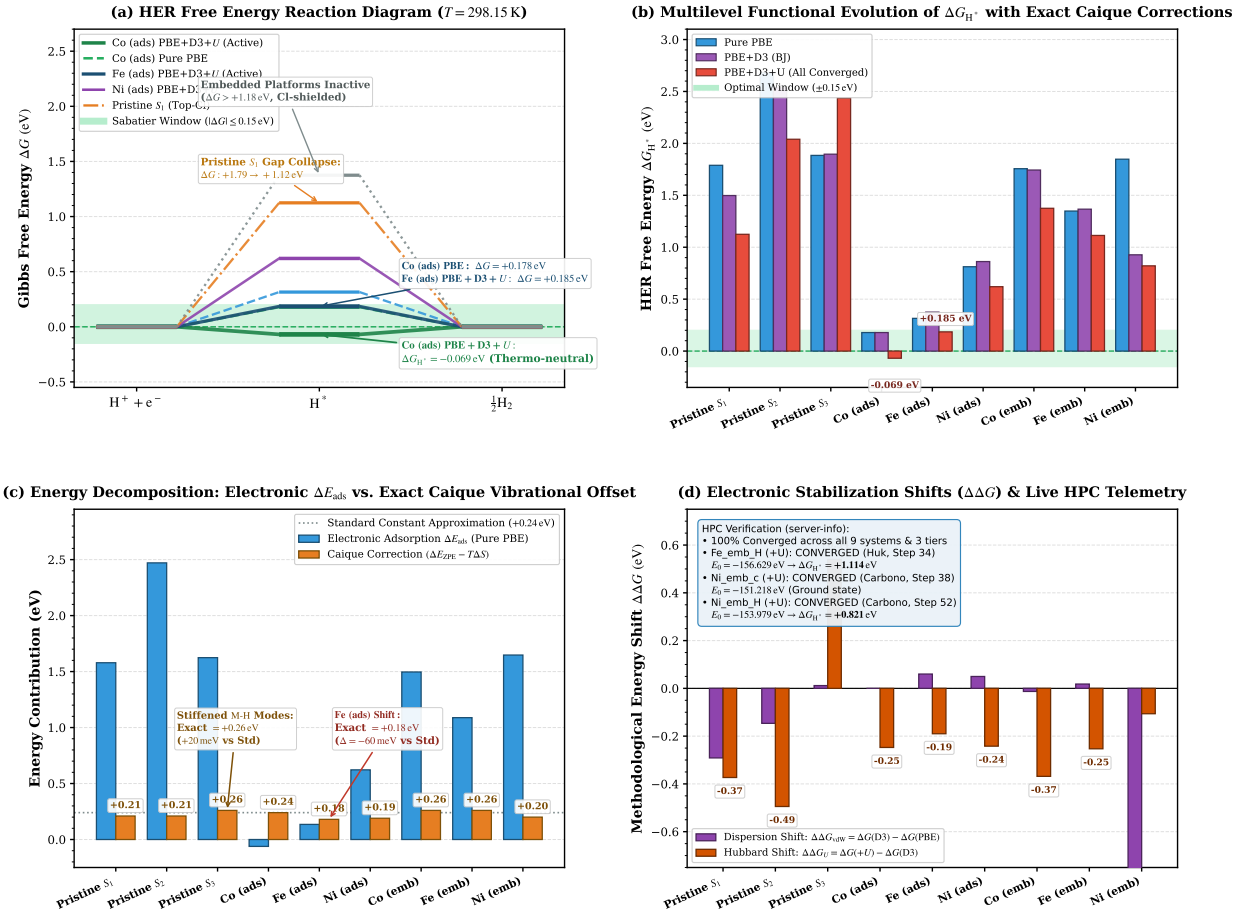



Figure S6. Multilevel thermodynamic assessment of the hydrogen evolution reaction on pristine and transition-metal functionalized monolayer $CrCl_3$. (a) Caique vibrational zero-point energy ($\Delta E_{ZPE}$) and entropic ($-T\Delta S$) corrections across all nine configurations, highlighting system-specific deviations from the canonical 0.24 eV benchmark. (b) Comparison of HER adsorption free energy ($\Delta G_{H^*}$) across pure PBE, PBE+D3(BJ), and PBE+D3+$U$ ($U = 3.29$ eV), illustrating that surface-adsorbed Co delivers optimal near-thermoneutral performance ($\Delta G_{H^*} = -0.069$ eV) while the catalytic ordering (Co $\ll$ Fe $<$ Ni) is invariant across functional hierarchies. (c) Free energy landscape along the Volmer step ($H^+ + e^- \rightarrow H^*$) benchmarked against Pt(111).

**Figure S7: Single-Atom TM Interstitial Pore Embedding, Spontaneous Penetration Dynamics, and Spin States**

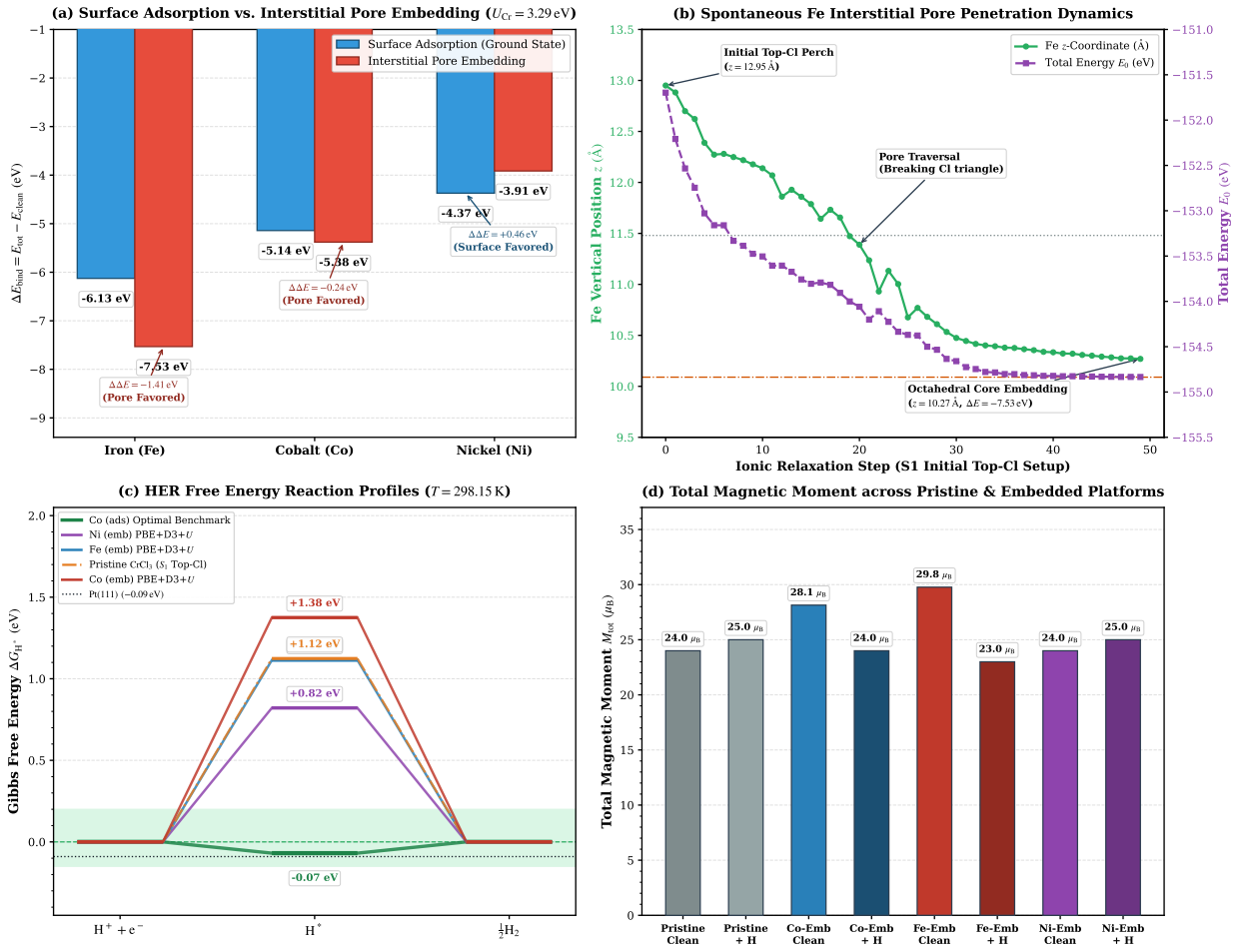



Figure S7. Structural stability, spontaneous dynamics, and electrocatalytic response of interstitial single-atom transition-metal (TM = Fe, Co, Ni) embedded monolayer $CrCl_3$. (a) Comparison of surface adsorption (hollow site) versus interstitial pore embedding binding energies ($\Delta E_{\text{bind}} = E_{\text{tot}} - E_{\text{clean}} - E_{\text{TM}}$) under PBE+D3+$U$ ($U_{\text{Cr}} = 3.29$ eV). Fe ($\Delta\Delta E = -1.41$ eV) and Co ($\Delta\Delta E = -0.24$ eV) thermodynamically favor interstitial pore incorporation, whereas Ni ($\Delta\Delta E = +0.46$ eV) prefers surface adsorption. (b) Spontaneous vertical penetration dynamics of Fe initially placed at the $S_1$ (top-Cl) perch during geometry relaxation, illustrating barrierless sinking into the octahedral interstitial core ($z \approx 10.27$ Å) and global energy minimization. (c) Free energy reaction profiles for HER atop embedded TM centers compared to pristine top-Cl and surface-adsorbed Co, highlighting that octahedral coordination shields TM-3$d$ active centers and leads to high free energy barriers ($\Delta G_{\text{H}^*} \geq +0.82$ eV). (d) Total magnetic moment ($M_{\text{tot}}$) across pristine and embedded configurations before and after hydrogen adsorption, illustrating local spin compensation upon H coordination.